

Introduction to computing

Dictionaries:- ^{→ it is} a data structure



it has no order, list is also a data structure but it has order.

Exmp list:-

lst = [1, 2, 3, 4]
0 1 2 3 → index

white space

indentation

tab

space

Exmp dictionary:-

d = {

"package": "A package is a module that
can be added to any program",

"random": "Any information or data that has
no order",

"unary operator": "An operator that take one
value"
}

2>

So d is the variable that take this dictionary.

d['package']

A package ... program

"random": ["first meaning", "second meaning"]
→ d['random'] it give the thing inside it

d['python']
↓

key error because it not exist in our dictionary.

Iterating over dictionaries

```
for a in d:  
    print a
```

↓

package

random

unary operator

```
for a in d
```

```
    v = d[a]
```

```
    print a, "=", v
```

package = ---

random = ---

unary op = ---

Is key exist in dictionary?

'package' in d

True

'else' in d

False

`d.items()` # returns a list of tuple and each tuple has a key and a value.

```
{[('package', '—'), (random, '—'), ( )]}
```

↓ ↓
key value

```
( for k, v in d.items():  
  print k, "=", v ) → v-imp syntax
```

Using dict for configuration

```
autograd = {  
  'url' : '121-52-146-108:8000',  
  'user' : 'p1111',  
  'subpas' : 'abc-334'  
}
```


agar ap ko order data ha to list
is best if you know only the key name
than you will use dictionary.

e.g num_name = {
 1: "one",
 2: "two",
 3: "three"
 20: "twenty"
 1000: "thousand"
}

we use string and
tuple in key that
not change.

num_names[20]

'twenty'

↳ not use list because we
can use like 1000, 1
but in list we have
to remember the order
number.

num_names [1, "1" string]

'one'

called data type casting

Reading assignment:

composing program 2.4.3

In C the dictionary is called #table

No SQL = collection of dictionary